Microstates and Macroscopic Thermodynamics of a Binary System

Statistical Physics – Laboratory Practice 1

Section 2 / Group E

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Academic year 2026/2027
November 4, 2026

Abstract

Independent unbiased coins provide an exactly solvable two-state model. We enumerate the 128 microstates of seven coins, compute exact multiplicities and log-space distributions, and measure computational scaling. We derive finite-system probabilities, Boltzmann entropy changes and waiting times, then distinguish concentration of the heads fraction from the decreasing probability of any single macrostate in the thermodynamic limit. Automated checks compare the implementation with independent exact combinatorial values. The group suffixes give $M = 31.20656885$; nearest-integer rounding gives $N = 131$.

Repository

`https://github.com/martinlagunamolina-commits/Laboratory-1-statistical-physics`

1 Introduction & Objectives

The objective is to connect microstate counting with macrostate probabilities, Boltzmann entropy and the thermodynamic limit. We enumerate a small system, measure computational scaling, and calculate the group's probabilities, entropy differences and waiting times.

Each coin is independent with $P(H) = P(T) = 1/2$. A microstate specifies the ordered outcomes of all N coins; the macrostate $K = k$ specifies only the heads count. Thus there are 2^N equiprobable microstates and $N + 1$ macrostates:

$$\Omega(N,k) = \binom{N}{k} = \frac{N!}{k!(N-k)!}, \qquad P_k = 2^{-N}\Omega(N,k), \qquad \sum_{k=0}^{N} \Omega(N,k) = 2^N. \tag{1}$$

These are discrete sums, not exact integrals of a smoothed curve.

2 Methodology & Collaborative Workflow

2.1 Environment & Reproducibility

The project uses the `statphys_lab1` Conda environment specified in `environment.yml`. The submitted Windows execution log records Python 3.11.18, NumPy 2.4.6, SciPy 1.17.1 and Matplotlib 3.11.2. The resolved package specification is exported as `environment-resolved.yml` for reproduction on the team laptops. Exact counting has no sampling uncertainty. Floating-point results are checked against exact coefficients with the tolerances defined in the tests. Five repeated calibration runs estimate timing variability for their workload.

2.2 Version Control & Repository Architecture

The repository separates `src/`, `tests/`, `results/` and `report/`. The entry point `main.py` exports numerical data, PDF vector figures and LaTeX fragments. The benchmark script records elapsed time and peak resident memory in isolated processes. The existing Git history contains feature branches and merged pull requests. The delivery workflow requires a teammate's approval before merging final changes into `main`, followed by the `v1.0` release tag. The automated suite checks enumeration, binomial coefficients, log-gamma, normalization, symmetry, moments, entropy signs, waiting times and output generation.

3 Computational Analysis of Microstates

3.1 Enumeration for $N = 7$ (1.a)

1.a: Seven coins. For $N = 7$, $2^7 = 128$. All 128 configurations are listed by macrostate in Appendix B and exported to `results/microstates_N7.csv`. The counts are exact and carry no sampling uncertainty.

Table 1: Exact multiplicities for seven coins.

Heads k	0	1	2	3	4	5	6	7
Tails $7-k$	7	6	5	4	3	2	1	0
$\Omega(7,k)$	1	7	21	35	35	21	7	1

3.2 Full Enumeration Wall (1.b)

Integer s encodes an outcome through its N bits. The generator visits $s = 0, \ldots, (1 \ll N) - 1$ and yields one integer at a time. The timing workload consumes every yielded configuration and counts it, without storing, formatting or printing the entire list. It therefore measures full integer enumeration, not text-file writing or classification by heads. Formatting all outcomes additionally costs $O(N2^N)$ character operations. The enumeration loop requires $O(2^N)$ iterations and $O(N)$ live bits; calling `list()` would instead require exponentially increasing memory. Python's arbitrary-precision integers avoid fixed-width overflow but do not remove this cost.

Table 2: Full enumeration runtime and peak resident memory.

N	Time (s)	Peak RSS (MiB)	Status
20	0.108	17.72	complete
21	0.241	17.76	complete
22	0.416	17.85	complete
23	0.806	17.81	complete
24	1.495	17.74	complete
25	2.942	17.82	complete
26	5.953	17.77	complete
27	11.474	17.68	complete
28	22.757	17.81	complete
29	46.124	17.61	complete
30	91.921	17.73	complete
31	> 300	—	timeout

Largest completed tested candidate: $N = 30$. Per-run wall-clock budget: 300 s.

These measurements are from the recorded platform; consult the accompanying benchmark JSON for proof.

The measurements are recorded in the submitted JSON logs. The largest completed candidate is host-specific. Peak RSS includes the interpreter. Each large-N timing is one observation; repeated $N = 24$ calibration estimates variability for that workload only. A timeout records failure to complete within 300 seconds, not an extrapolated execution time.

3.3 Macrostate Frequency Distributions (1.c)

Exact coefficients are generated by $\Omega(N, 0) = 1$ and $\Omega(N, k+1) = \Omega(N, k)(N-k)/(k+1)$, with exact integer division. CSV files contain every value for $N = 20, 50, 80, 100$. The single left-hand plot overlays the four required raw multiplicity curves on a logarithmic vertical axis; normalized probabilities are also shown.

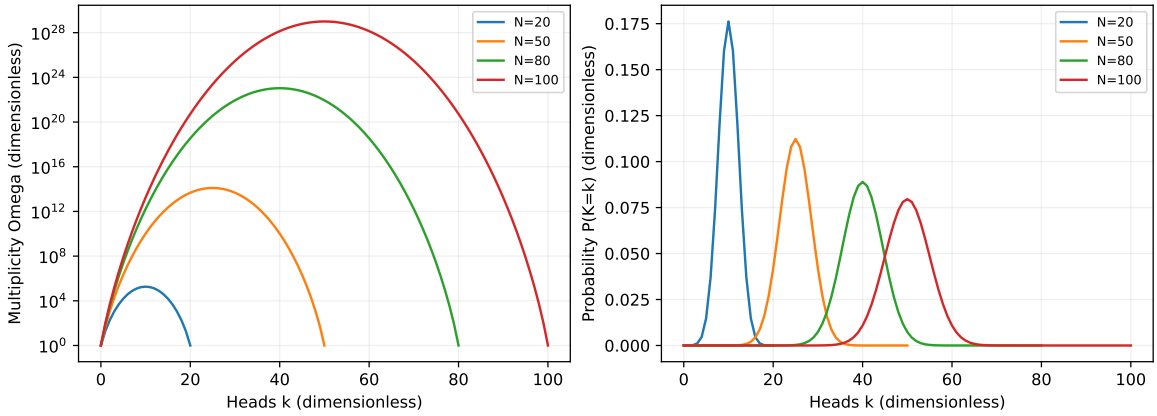


Figure 1: Multiplicity (left, logarithmic scale) and normalized probability (right) for $N = 20, 50, 80, 100$.

All axes are dimensionless. These deterministic curves have no sampling error bars; floating-point errors are assessed through independent exact checks.

3.4 Practical Distribution Limit (1.d)

Full enumeration is unnecessary:

$$\ln \Omega(N, k) = \ln \Gamma(N+1) - \ln \Gamma(k+1) - \ln \Gamma(N-k+1). \tag{2}$$

Logarithms prevent coefficient overflow, although subtracting large log-gamma values can lose precision. Binary64 exponentiation overflows near $\ln \Omega > 709.78$, reached by central coefficients at approximately $N = 1030$. Two retained float64 arrays require $16(N+1)$ bytes; plotting and temporaries need additional memory. The full-grid algorithm costs $O(N)$ time and memory. We measure calculation and vector-PDF plotting with a chosen 30-s timeout and 256-MiB peak-RSS acceptance budget, checked after execution rather than enforced as an operating-system memory cap.

Table 3: Full-grid distribution calculation and vector-PDF plotting.

N	Time (s)	Peak RSS (MiB)	Status
1000	2.346	90.96	complete
10000	2.090	91.29	complete
100000	1.970	95.27	complete
1000000	2.062	147.78	complete
2000000	2.225	218.07	complete
4000000	2.694	359.35	over_ram_budget

Largest completed tested candidate: $N = 2000000$. Per-run wall-clock budget: 30 s. Peak-RSS budget: 256 MiB.
These measurements are from the recorded platform; consult the accompanying benchmark JSON for provenance.

This is the maximum among the tested candidates under declared resource budgets, not an exhaustive or hardware-independent maximum. Sparse evaluation and analytic approximations permit much larger system sizes.

3.5 Thermodynamic Limit (1.e)

$\mathbb{E}K = N/2$, $\operatorname{Var} K = N/4$, $\sigma_K = \sqrt{N}/2$. For the heads fraction $X = K/N$, $\sigma_X = 1/(2\sqrt{N})$ and $\sigma_K/\mathbb{E}K = N^{-1/2}$. At $N = 10^{23}$, $\sigma_K = 1.58114 \times 10^{11}$ coins but $\sigma_X = 1.58114 \times 10^{-12}$. The fraction is sharply concentrated at $1/2$; the absolute width in coins increases. Chebyshev gives $P(|X - 1/2| \geq \epsilon) \leq 1/(4N\epsilon^2)$ for fixed $\epsilon > 0$.

4 Probabilities, Entropy, and Thermodynamics

4.1 Parameter Assignment

For r team members with two-digit NIA suffixes d_i,

$$M = \frac{r}{\sum_{i=1}^{r} 1/d_i}, \qquad N = 100 + \mathrm{round}_{\text{half-up}}(M). \tag{3}$$

NIA suffixes: 71, 95, 11, 79. $M = 5861405/187826 = 31.20656885$; $N = 131$ (nearest rounding); $k_0 = 65$. For 2.i, $(q_-, q_+) = (21, 41)$. Rounding is provisional because the manual does not specify it.

The nominal assignment $100 + M$ is noninteger for this group, so we adopt nearest-integer rounding. This convention requires instructor confirmation. For odd N, both 65 and 66 are modes; we use $k_0 = 65$ for individual-state questions and offsets. Interval bounds retain their literal half-integer center.

4.2 Probability and Entropy Analysis (2.a–g)

2.a–d: Probabilities. Equation (1) supplies the point probabilities. For an inclusive interval,

$$P(k_1 \le K \le k_2) = \sum_{k=k_1}^{k_2} \exp[\ln \Omega(N,k) - N \ln 2]. \tag{4}$$

The implementation uses log-sum-exp for the summation. If the number of tails $L = N - K$ is between $N/2 - 20$ and $N/2 + 20$, the heads count lies in the same symmetric interval. This is why 2.d can use the heads distribution.

Table 4: Group probabilities, entropy differences, waiting time and likelihood ratios for $N = 131$.

Part	Quantity	Result
2.a	$P(K = 30)$	1.24466×10^{-10}
2.b	$P(K = k_0)$	6.93155×10^{-2}
2.c	$P(\lvert K - N/2\rvert \le 10)$	9.19830×10^{-1}
2.d	$P(\lvert(N-K) - N/2\rvert \le 20)$	9.99569×10^{-1}
2.e	$\Delta S/k_B$	-9.31202×10^{0}
2.f	$\Delta S/k_B$	-1.36837×10^{0}
2.g	$\Delta S/k_B$	3.44931×10^{1}
2.h	$\mathbb{E}T$ (s)	1.57373×10^{27}
2.i	R_-, R_+	$1.80741 \times 10^{14},\ 1.10703 \times 10^{4}$
2.e	ΔS ($\mathrm{J\,K^{-1}}$)	-1.28566×10^{-22}
2.f	ΔS ($\mathrm{J\,K^{-1}}$)	-1.88924×10^{-23}
2.g	ΔS ($\mathrm{J\,K^{-1}}$)	4.76229×10^{-22}

All probabilities and $\Delta S/k_\mathrm{B}$ are dimensionless. Waiting times are in seconds. The numerical table is generated from `group_results.json`; the group uses $N = 131$, $k_0 = 65$, interval $56 \le K \le 75$ in 2.c, and $46 \le K \le 85$ in 2.d.

2.e–g: Entropy and its sign. Boltzmann's macrostate entropy is

$$S(k) = k_\mathrm{B} \ln \Omega(N,k), \qquad \Delta S_{k_1 k_2} = S(k_1) - S(k_2) = k_\mathrm{B} \ln \frac{\Omega(N,k_1)}{\Omega(N,k_2)}, \tag{5}$$

where $k_\mathrm{B} = 1.380649 \times 10^{-23}\,\mathrm{J\,K^{-1}}$ exactly in SI. Therefore 2.e and 2.f are negative: offset states have fewer microstates than the mode. Part 2.g is positive because the mode has more microstates than $k = 20$. The signs follow the manual's definition, which is the reverse of "final minus initial" if $k_1 \to k_2$ is read as a temporal transition. The labels "high" or "moderate disorder" are qualitative; the actual ordering follows the computed multiplicities.

4.3 Waiting Times and Likelihood (2.h–i)

2.h: Eight heads. At N individual coins per second, one independent experiment tossing all N coins takes one second. For success probability $p_8 = \binom{N}{8}2^{-N}$ and independent complete trials, the first-success count T is geometric on $1, 2, \ldots$:

$$P(T = t) = (1 - p_8)^{t-1} p_8, \qquad \mathbb{E}T = \frac{1}{p_8} \text{ s}, \qquad \sigma_T = \frac{\sqrt{1 - p_8}}{p_8} \text{ s}. \tag{6}$$

Thus a huge mean waiting time is not a deterministic deadline. For rare events the standard deviation is approximately the mean. After t complete trials, $P(T \le t) = 1 - (1 - p_8)^t$. At a rate f complete trials/s, both the mean and standard deviation in seconds are divided by f. If "N coins/s" instead means sequentially replacing single coins, trials are correlated and this geometric model is not justified without further dynamics. The program retains $\log_{10} \mathbb{E}T$ to avoid overflow and also exports years, using 365.25×86400 s/year as the conversion convention.

2.i: Likelihood ratio. The printed manual says $M \pm 10$ heads, not $N/2 \pm 10$. We evaluate the literal two cases:

$$R_\pm = \frac{P(K = k_0)}{P(K = q_\pm)} = \exp[\ln \Omega(N, k_0) - \ln \Omega(N, q_\pm)], \qquad q_\pm = \text{round}_{\text{half-up}}(M \pm 10). \tag{7}$$

For this group, $q_- = 21$ and $q_+ = 41$. The same nearest-integer rounding convention is used for these states. Both ratios exceed one because k_0 is a mode.

4.4 Second Law of Thermodynamics Interpretation (2.j)

2.j: Statistical Second Law. For finite N, a lower-multiplicity macrostate has nonzero probability. Rare entropy decreases therefore do not contradict the statistical interpretation of the Second Law. They are overwhelmingly unlikely for a macroscopic entropy decrease, not forbidden by the counting model. Macrostates are not equiprobable: microstates are.

Coin tossing also involves an external drive. This counting experiment alone does not specify an isolated system's dynamics, energy exchanges or total entropy production, so it cannot establish a deterministic entropy-increase trajectory. It shows why equilibrium macrostates dominate when microstates are sampled without bias. Individual ordered outcomes remain possible even though almost every observed macroscopic fraction lies very near one half. The maximal Boltzmann macrostate entropy must also be distinguished from the Shannon/Gibbs entropy $Nk_\text{B} \ln 2$ of the uniform ensemble over all microstates.

Numerical reliability. Exact integer checks establish multiplicities without sampling error. Floating-point comparisons use stated tolerances in the tests, not statistical confidence intervals. Six-significant-figure display is a presentation choice. Benchmarks are subject to host load and are reported as observations; repeated measurements on the target laptop are required to estimate their variability.

4.5 Macroscopic Re-evaluation (2.k)

For even N and $k = N/2 + \delta$ near the center, Stirling's formula yields

$$P_k \simeq \sqrt{\frac{2}{\pi N}} e^{-2\delta^2/N}, \quad \ln \Omega(N,k) \simeq N \ln 2 - \tfrac{1}{2} \ln(\pi N/2) - 2\delta^2/N. \tag{8}$$

This local approximation is not appropriate for fixed small k near an edge. There, for fixed m, $\binom{N}{m} \sim N^m/m!$. For fixed fraction $x = k/N$ in $(0,1)$,

$$\ln \Omega(N, Nx) \simeq N h(x) - \tfrac{1}{2} \ln[2\pi N x(1-x)], \quad h(x) = -x \ln x - (1-x)\ln(1-x). \tag{9}$$

Hence $P(K \simeq Nx)$ is exponentially suppressed away from $x = 1/2$, with rate $\ln 2 - h(x) > 0$. This gives the following separate answers to 2.a–j.

Table 5: Macroscopic limits of the finite-system quantities.

Part	Limit as $N \to \infty$ (fixed integer offsets)
2.a	$P_{30} \sim N^{30} 2^{-N}/30! \to 0$; use its logarithm for numerical representation.
2.b	$P_{N/2} \sim \sqrt{2/(\pi N)} \to 0$, despite remaining a most probable single macrostate.
2.c	$P(\lvert K - N/2 \rvert \le 10) \sim 21\sqrt{2/(\pi N)} \to 0$ for even N.
2.d	The symmetric tails interval has probability $\sim 41\sqrt{2/(\pi N)} \to 0$.
2.e	$\Delta S/k_{\mathrm{B}} \sim -2(25)^2/N = -1250/N \to 0^-$.
2.f	$\Delta S/k_{\mathrm{B}} \sim -2(10)^2/N = -200/N \to 0^-$.
2.g	$\Delta S/k_{\mathrm{B}} \simeq N \ln 2 - \tfrac{1}{2}\ln(\pi N/2) - 20 \ln N + \ln(20!) \to +\infty$.
2.h	$\mathbb{E}T \sim 8!\, 2^N/N^8$ s $\to \infty$; $\sigma_T/\mathbb{E}T \to 1$.
2.i	For fixed valid $q_\pm = M \pm 10 = m$, $R_\pm \sim m!\, 2^N \sqrt{2/(\pi N)}/N^m \to \infty$.
2.j	A fixed fractional imbalance becomes exponentially improbable. Finite-system fluctuations remain consistent with a statistical Second Law.

For odd N, the literal intervals in 2.c–d contain 20 and 40 states instead of 21 and 41. At $N = 10^{23}$ (even), $P_{N/2} \simeq 2.52313 \times 10^{-12}$; the two fixed-width interval probabilities are approximately 5.29858×10^{-11} and 1.03448×10^{-10}. Part 2.e gives $\Delta S \simeq -1.72581 \times 10^{-43}$ J/K, and 2.f gives -2.76130×10^{-44} J/K. Part 2.g has leading value $\Delta S \simeq 0.956993$ J/K. For 2.h, $\log_{10}(\mathbb{E}T/\mathrm{s}) \simeq N \log_{10} 2 - 8 \log_{10} N + \log_{10}(8!)$, whose leading term is 3.01030×10^{22}.

In this extrapolation, N varies independently of the finite-group assignment $100 + M$, while the group's M and rounded $q_\pm$ remain fixed.

Fixed-width intervals shrink relative to σ_K; this explains their vanishing probability. An interval of fixed width in *fraction* K/N has probability tending to one. Direct log-gamma subtraction at 10^{23} cannot resolve small offsets in binary64; equations above are used analytically.

5 Conclusion

The binomial model connects exact microstate counting with macrostate probabilities and Boltzmann entropy. Streaming enumeration limits live memory, but its runtime grows exponentially. Log-space combinatorics avoids coefficient overflow and supports much larger distributions. At macroscopic sizes, the heads fraction concentrates near one half even though the probability of each individual central macrostate decreases. Rare entropy decreases remain consistent with the statistical interpretation of the Second Law.

A AI Collaboration & Audit Statement

In compliance with course policy, AI tools were utilized as code assistants. All generated algorithms were audited and validated using automated tests (pytest). Prompt logs and verification steps are recorded in AI_STATEMENT.md in the repository.

B All seven-coin microstates

H denotes heads and T tails. Every string occurs exactly once. The exact macrostate counts sum to 128. The CSV export contains the same complete list.

0 heads, 7 tails; $\Omega = 1$**:**

```
TTTTTTT
```

1 heads, 6 tails; $\Omega = 7$**:**

```
TTTTTTH   TTTTTHT   TTTTHTT   TTTHTTT   TTHTTTT   THTTTTT   HTTTTTT
```

2 heads, 5 tails; $\Omega = 21$**:**

```
TTTTTHH   TTTTHTH   TTTTHHT   TTTHTTH   TTTHTHT   TTTHHTT   TTHTTTH   TTHTTHT
TTHTHTT   TTHHTTT   THTTTTH   THTTTHT   THTTHTT   THTHTTT   THHTTTT   HTTTTTH
HTTTTHT   HTTTHTT   HTTHTTT   HTHTTTT   HHTTTTT
```

3 heads, 4 tails; $\Omega = 35$**:**

```
TTTTHHH   TTTHTHH   TTTHHTH   TTTHHHT   TTHTTHH   TTHTHTH   TTHTHHT   TTHHTTH
TTHHTHT   TTHHHTT   THTTTHH   THTTHTH   THTTHHT   THTHTTH   THTHTHT   THTHHTT
THHTTTH   THHTTHT   THHTHTT   THHHTTT   HTTTTHH   HTTTHTH   HTTTHHT   HTTHTTH
HTTHTHT   HTTHHTT   HTHTTTH   HTHTTHT   HTHTHTT   HTHHTTT   HHTTTTH   HHTTTHT
HHTTHTT   HHTHTTT   HHHTTTT
```

4 heads, 3 tails; $\Omega = 35$**:**

```
TTTHHHH   TTHTHHH   TTHHTHH   TTHHHTH   TTHHHHT   THTTHHH   THTHTHH   THTHHTH
THTHHHT   THHTTHH   THHTHTH   THHTHHT   THHHTTH   THHHTHT   THHHHTT   HTTTHHH
HTTHTHH   HTTHHTH   HTTHHHT   HTHTTHH   HTHTHTH   HTHTHHT   HTHHTTH   HTHHTHT
HTHHHTT   HHTTTHH   HHTTHTH   HHTTHHT   HHTHTTH   HHTHTHT   HHTHHTT   HHHTTTH
HHHTTHT   HHHTHTT   HHHHTTT
```

5 heads, 2 tails; $\Omega = 21$**:**

```
TTHHHHH   THTHHHH   THHTHHH   THHHTHH   THHHHTH   THHHHHT   HTTHHHH   HTHTHHH
HTHHTHH   HTHHHTH   HTHHHHT   HHTTHHH   HHTHTHH   HHTHHTH   HHTHHHT   HHHTTHH
HHHTHTH   HHHTHHT   HHHHTTH   HHHHTHT   HHHHHTT
```

6 heads, 1 tails; $\Omega = 7$**:**

```
THHHHHH   HTHHHHH   HHTHHHH   HHHTHHH   HHHHTHH   HHHHHTH   HHHHHHT
```

7 heads, 0 tails; $\Omega = 1$**:**

```
HHHHHHH
```